



C23-EC-403

23432

BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

DECE – FOURTH SEMESTER EXAMINATION

MICROWAVE AND SATELLITE COMMUNICATION SYSTEMS

Time : 3 hours]

[Total Marks : 80

PART—A

3×10=30

- Instructions :**
- (1) Answer **all** questions.
 - (2) Each question carries **three** marks.
 - (3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

- 1. Define reflection coefficient and standing wave ratio.
- 2. List the limitations of ground wave propagation.
- 3. List the applications of smart antennas.
- 4. Define beam width of an antenna.
- 5. Define dominant mode and cut off wave length of waveguide.
- 6. List different microwave solid state devices.
- 7. Explain the basic principle of RADAR.
- 8. What is the need of Duplexer?
- 9. What is the function of transponder?
- 10. List the advantages of geosynchronous satellites.

PART—B

10×5=50

- Instructions :** (1) Answer *any five* questions.
(2) Each question carries **ten** marks.
(3) Answers should be comprehensive and the criteria for valuation is the content but not the length of the answer.

- 11.** Explain the space wave propagation and factors affecting space wave propagation.
- 12.** (a) Classify different layers of ionosphere and briefly explain them. 5
(b) Explain Horn antennas and give their applications. 5
- 13.** Explain the operation of broad side array and end fire array with radiation pattern. 5+5
- 14.** Explain the working principle of reflex klystron.
- 15.** Explain the working of IMPATT diode.
- 16.** Draw the block diagram of Pulse RADAR and explain the function of each block.
- 17.** Derive the RADAR range equation.
- 18.** Draw and explain block diagram of satellite communication system.

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